

The first smooth-absolute IRMF component:
concentration, a scalar expansion, and Gaussian error bounds

4 Theoretical guarantees

The first IRMF component fits a local constant to noisy observations using Gaussian-smoothed absolute loss. This note separates its error into a population bias, a leading random term, and a remainder. The general expansion needs independent errors with finite first moments. Gaussian noise is needed only for the explicit curvature and bias formulas in the later results.

Main result and reading guide

Write $\tilde{S}_t$ for the fit to observed data, S_t for the minimizer of the expected noisy loss, and X_t for the clean signal value. The expansion is

$$\boxed{\tilde{S}_t - X_t = \underbrace{S_t - X_t}_{\text{population bias}} + \underbrace{F_t^{-1}\nabla_t}_{\text{leading random term}} + \underbrace{R_t}_{\text{remainder}} .}$$

Here F_t is the population curvature at S_t , and ∇_t is the centered empirical score there. The identity defines R_t ; Proposition 4.1 bounds it. In particular, the target of the stochastic expansion is S_t , which generally differs from X_t . Subtracting $F_t^{-1}\nabla_t$ is an oracle correction, since it requires unknown population quantities.

For orientation, let n be the number of observations, m the number of evaluation points, h the window radius, and H the loss smoothing scale. Put $z_x = x + \log(4m)$, where $x \geq 0$ sets the failure probability e^{-x} . Under independent Gaussian noise, fixed $H > 0$ and noise scale, regular symmetric windows, bounded first and second signal derivatives, and normalized population curvature bounded away from zero, the bounds below give

$$\max_t |\tilde{S}_t - X_t| \lesssim \underbrace{h^2}_{\text{bias}} + \underbrace{\sqrt{z_x/(nh)}}_{\text{leading random term}} + \underbrace{z_x/(nh)}_{\text{remainder}}$$

with probability at least $1 - e^{-x}$, once the stated local sample-size conditions hold. The symbol $\lesssim$ hides constants fixed by these assumptions. This rate summary is made explicit in Corollaries 4.3–4.4 and Proposition 4.5.

The main text gives definitions, assumptions, and results. All proofs are in Appendix A. Proposition 4.1 and Lemma 4.2 retain the numbering of the original IMF draft.

4.1 Model and general expansion

Let u range over n deterministic observation locations in $[0, 1]$. Let $\mathcal{T}_n \subset [0, 1]$ be a nonempty finite set of evaluation points and put $m = |\mathcal{T}_n|$. The signal, design, bandwidth, and weights are deterministic. For $t \in \mathcal{T}_n$, write

$$w_{u,t} := K_h(t - u), \quad W_t := \sum_u w_{u,t}, \quad Q_t^2 := \sum_u w_{u,t}^2, \quad k_t := \max_u w_{u,t}. \quad (1)$$

Assume $w_{u,t} \geq 0$ and $W_t > 0$. Suppose that $Y_u = X_u + \varepsilon_u$, where the errors are independent and identically distributed with $\mathbb{E}|\varepsilon_u| < \infty$, and let

$$\phi(z) := \mathbb{E}\rho(z + \varepsilon). \quad (2)$$

The argument below also allows independent, non-identically distributed errors with finite first moments if $\phi(X_u - x)$ in each summand is replaced by $\mathbb{E}\rho(Y_u - x)$.

Assume that $\rho \in C^3(\mathbb{R})$ is convex and coercive, meaning $\rho(z) \rightarrow \infty$ as $|z| \rightarrow \infty$, and, for finite constants L_1, L_2, L_3 ,

$$|\rho'(z)| \leq L_1, \quad 0 \leq \rho''(z) \leq L_2, \quad |\rho'''(z)| \leq L_3, \quad z \in \mathbb{R}. \quad (3)$$

For the Gaussian-smoothed absolute-value contrast, with a deterministic $H > 0$,

$$\begin{aligned} \rho_H(r) &:= \mathbb{E}_Z |r + HZ|, \quad Z \sim N(0, 1), \\ \rho'_H(r) &= \operatorname{erf}\left(\frac{r}{\sqrt{2}H}\right), \quad \rho''_H(r) = \frac{\sqrt{2/\pi}}{H} e^{-r^2/(2H^2)}, \\ \rho'''_H(r) &= -\frac{\sqrt{2/\pi} r}{H^3} e^{-r^2/(2H^2)}. \end{aligned} \quad (4)$$

One may take the conservative constants

$$L_1 = 1, \quad L_2 = \frac{2}{H\sqrt{2\pi}}, \quad L_3 = \frac{2}{H^2\sqrt{2\pi}}. \quad (5)$$

The Lipschitz bound $|\rho'| \leq L_1$ and the finite first noise moment make the expected objectives finite. The inequality $\mathbb{E}\rho(Y_u - z) \geq \rho(-z) - L_1 \mathbb{E}|Y_u|$ also makes each population objective coercive. Thus both empirical and population minimizers exist. Define the empirical fit $\tilde{S}_t$, its population target S_t , the population curvature F_t , and the centered score ∇_t by

$$\tilde{S}_t := \arg \inf_{z \in \mathbb{R}} \sum_u \rho(Y_u - z) w_{u,t}, \quad S_t := \arg \inf_{z \in \mathbb{R}} \sum_u \mathbb{E}\rho(Y_u - z) w_{u,t}, \quad (6)$$

$$F_t := \sum_u \phi''(X_u - S_t) w_{u,t}, \quad D_t := F_t^{1/2}, \quad (7)$$

$$\nabla_t := \sum_u \{\rho'(Y_u - S_t) - \mathbb{E}\rho'(Y_u - S_t)\} w_{u,t}. \quad (8)$$

Require $F_t > 0$ for every $t \in \mathcal{T}_n$; convexity then makes the population minimizer unique. For ρ_H , both objectives are strictly convex and $F_t > 0$ whenever $W_t > 0$. They need not be globally strongly convex: their curvatures tend to zero in the tails. The proof uses positive curvature near S_t , not a uniform curvature lower bound on $\mathbb{R}$.

The target S_t minimizes the expected noisy loss. It generally differs from X_t and from the minimizer of $\sum_u w_{u,t} \rho_H(X_u - z)$.

The constants below separate three controls: C_1 bounds the centered score, C_2 bounds the curvature fluctuation, and C_3 bounds the third derivative. The scale $D_t = \sqrt{F_t}$ expresses the first two bounds in local curvature units. For $x \geq 0$ and $z_x := x + \log(4m)$, set

$$C_1 := \max_{t \in \mathcal{T}_n} \frac{L_1}{D_t} \left\{ Q_t \sqrt{2z_x} + \frac{2k_t z_x}{3} \right\}, \quad (9)$$

$$C_2 := \max_{t \in \mathcal{T}_n} \frac{L_2}{D_t} \left\{ Q_t \sqrt{2z_x} + \frac{k_t z_x}{3} \right\}, \quad (10)$$

$$C_3 := \max_{t \in \mathcal{T}_n} \frac{L_3 W_t}{D_t^2}, \quad D_{\min} := \min_{t \in \mathcal{T}_n} D_t, \quad \delta := \frac{C_2}{D_{\min}}. \quad (11)$$

The parameter δ bounds the relative curvature error. The two conditions below keep the empirical curvature positive near S_t and make the scalar Taylor expansion valid at the empirical minimizer.

Proposition 4.1. *Suppose that the conditions above hold and that*

$$\delta < 1, \quad \frac{C_1 C_3}{D_{\min}(1 - \delta)^2} < \frac{4}{9}. \quad (12)$$

Then there is a random set $\Omega(x)$ with $\mathbb{P}\{\Omega(x)\} \geq 1 - e^{-x}$ such that

$$\sup_{t \in \mathcal{T}_n} \left| D_t (\tilde{S}_t - S_t - F_t^{-1} \nabla_t) \right| \leq \frac{1}{D_{\min}} \left\{ \frac{9C_1^2 C_3}{8(1 - \delta)^3} + \frac{C_1 C_2}{1 - \delta} \right\}. \quad (13)$$

On the same event, every empirical minimizer is unique and

$$\sup_{t \in \mathcal{T}_n} D_t |\tilde{S}_t - S_t| \leq \frac{3C_1}{2(1 - \delta)}. \quad (14)$$

Consequently, the oracle-corrected estimator

$$\tilde{S}_t^{\text{corr}} := \tilde{S}_t - F_t^{-1} \nabla_t \quad (15)$$

satisfies the same upper bound with $\tilde{S}_t - S_t - F_t^{-1} \nabla_t = \tilde{S}_t^{\text{corr}} - S_t$.

Remark (Sign of the correction). *With ∇_t defined by (8), the minus sign in (13) is forced by differentiation: $f'_o(S_t) = -\nabla_t$. Thus $\tilde{S}_t - S_t$ has first-order term $F_t^{-1} \nabla_t$, and one subtracts that term to correct $\tilde{S}_t$. The plus sign in the unfinished display in the original IMF draft would be correct only if ∇_t were defined with the opposite sign.*

4.2 Score and curvature bounds

The next lemma supplies the probabilistic input to Proposition 4.1. Its score and curvature bounds use Bernstein's inequality and a union bound over the finite evaluation grid. The third derivative bounds are deterministic.

Fix $t \in \mathcal{T}_n$. All quantities in this subsection may depend on t . For the proof, abbreviate the population and empirical objectives as

$$f(z) := \sum_u \phi(X_u - z) w_{u,t}, \quad f_o(z) := \sum_u \rho(Y_u - z) w_{u,t}, \quad (16)$$

and

$$x^* := \arg \inf_z f(z) = S_t, \quad x_o := \arg \inf_z f_o(z) = \tilde{S}_t. \quad (17)$$

Further, put

$$F_{\circ} := f''_{\circ}(x^*), \quad F := f''(x^*) = D^2. \quad (18)$$

For $r > 0$, let $U_t(r) := \{z \in \mathbb{R} : |D(z - x^*)| \leq r\}$.

Lemma 4.2. *There is an event $\Omega(x)$ of probability at least $1 - e^{-x}$ on which, simultaneously for every $t \in \mathcal{T}_n$,*

(i)

$$\frac{1}{D} |f'(x^*) - f'_{\circ}(x^*)| = \frac{1}{D} \left| \sum_u \{ \rho'(Y_u - x^*) - \mathbb{E} \rho'(Y_u - x^*) \} w_{u,t} \right| \leq C_1, \quad (19)$$

$$\frac{1}{D} |f''(x^*) - f''_{\circ}(x^*)| = \frac{1}{D} \left| \sum_u \{ \rho''(Y_u - x^*) - \mathbb{E} \rho''(Y_u - x^*) \} w_{u,t} \right| \leq C_2. \quad (20)$$

(ii)

$$\sup_{z \in \mathbb{R}} \frac{1}{D^2} \left| \sum_u \rho'''(Y_u - z) w_{u,t} \right| \leq C_3. \quad (21)$$

(iii) For every $r > 0$,

$$\sup_{z \in U_t(r)} \frac{1}{D^2} \left| \sum_u \{ \phi''(X_u - z) - \phi''(X_u - x^*) \} w_{u,t} \right| \leq \frac{C_3 r}{D}. \quad (22)$$

The deterministic argument in Appendix A.2 also applies to any other constants C_1, C_2, C_3 satisfying the bounds of Lemma 4.2 on a specified event. In particular, sharper variance bounds can be used for Gaussian noise.

4.3 Gaussian noise and normalized weights

Multiplying all weights in a window by a positive constant changes neither minimizer nor $F_t^{-1} \nabla_t$. Define

$$a_{u,t} := \frac{w_{u,t}}{W_t}, \quad N := \min_t \frac{1}{\sum_u a_{u,t}^2}, \quad \alpha := \max_{t,u} a_{u,t}. \quad (23)$$

Here N is the smallest effective sample size across windows. The population curvature and score for the normalized objective are $\bar{F}_t = F_t/W_t$ and $\bar{\nabla}_t = \nabla_t/W_t$. Their ratio is still $F_t^{-1} \nabla_t$.

Assume now $\varepsilon_u \sim N(0, \sigma^2)$ independently, with $\sigma \geq 0$, and $\rho = \rho_H$. Put

$$c := \sqrt{2/\pi}, \quad B := \sqrt{H^2 + \sigma^2}. \quad (24)$$

Independence of the Gaussian variables in the two expectations gives the exact identity

$$\phi(r) = \mathbb{E}_{\varepsilon} \rho_H(r + \varepsilon) = \rho_B(r). \quad (25)$$

Thus the population target is the clean-data fit at scale B , and

$$\sum_u a_{u,t} \operatorname{erf} \left(\frac{X_u - S_t}{\sqrt{2}B} \right) = 0, \quad \bar{F}_t = \frac{c}{B} \sum_u a_{u,t} e^{-(X_u - S_t)^2 / (2B^2)}. \quad (26)$$

The weighted score is positive below all active signal values and negative above them. Its unique zero S_t therefore lies between the smallest and largest X_u having positive weight. If

$$A \geq \max_t \max_{u,v: a_{u,t}a_{v,t} > 0} |X_u - X_v|, \quad \lambda := \frac{c}{B} e^{-A^2/(2B^2)}, \quad (27)$$

then $\bar{F}_t \geq \lambda > 0$ at every evaluation point. More generally, any known positive lower bound $\lambda \leq \min_t \bar{F}_t$ can be used below.

The bounds s_1 and s_2 below control the normalized score and curvature fluctuations. Dividing s_2 by the curvature lower bound λ gives the relative perturbation d . For $L_2 = c/H$ and $L_3 = c/H^2$, define

$$\begin{aligned} v_1 &:= \min\{1, \sigma L_2\}, & v_2 &:= \min\{L_2, \sigma L_3\}, \\ s_1 &:= v_1 \sqrt{\frac{2z_x}{N}} + \frac{2\alpha z_x}{3}, & s_2 &:= v_2 \sqrt{\frac{2z_x}{N}} + \frac{L_2 \alpha z_x}{3}, \\ d &:= \frac{s_2}{\lambda}, & z_x &= x + \log(4m). \end{aligned} \quad (28)$$

Corollary 4.3 (Explicit Gaussian bounds). *Under the Gaussian model and the deterministic design conditions above, suppose*

$$d < 1, \quad \frac{s_1 L_3}{\lambda^2(1-d)^2} < \frac{4}{9}. \quad (29)$$

With probability at least $1 - e^{-x}$, simultaneously for all t ,

$$|\tilde{S}_t - S_t| \leq \frac{3s_1}{2\lambda(1-d)}, \quad (30)$$

$$|F_t^{-1} \nabla_t| \leq \frac{s_1}{\lambda}, \quad (31)$$

$$|R_t| \leq \mathcal{R}_x := \frac{9L_3 s_1^2}{8\lambda^3(1-d)^3} + \frac{s_1 s_2}{\lambda^2(1-d)}, \quad (32)$$

$$R_t := \tilde{S}_t - S_t - F_t^{-1} \nabla_t.$$

In particular, $|\tilde{S}_t^{\text{corr}} - S_t| \leq \mathcal{R}_x$. All quantities in these bounds are specified by the weights, x , H , σ , and a curvature lower bound such as (27).

Remark (Dependence on the noise and loss scales). *The noise scale enters through $B = \sqrt{H^2 + \sigma^2}$, the lower bound λ , and the variance bounds v_1, v_2 . These are sufficient, conservative constants. At $\sigma = 0$, the empirical and population objectives coincide exactly, so both the estimation error and R_t vanish; the Bernstein linear terms need not vanish. If H varies with sample size, the explicit conditions must be checked again: L_2 and L_3 grow as H^{-1} and H^{-2} when H decreases.*

4.4 Bandwidth and sample-size conditions

Suppose that, for a known constant $C_K > 0$,

$$N \geq \frac{nh}{C_K}, \quad \alpha \leq \frac{C_K}{nh}. \quad (33)$$

These assumptions say that the effective sample size is at least a constant times nh , and no individual observation has disproportionate weight. They can be checked directly from the normalized weights.

For an equally spaced design, the calculation is clearest in terms of the actual grid spacing Δ . Use untruncated interior windows, or a periodic grid with each observation counted once, and take $0 < h < 1/2$. Let $w_{u,t} = K((t-u)/h)$, using local offsets in the periodic case, with K supported on $[-1, 1]$, $0 \leq K \leq K_{\max}$, and $K(v) \geq k_0 > 0$ on $|v| \leq \eta$, where $0 < \eta < 1$. If $\eta h \geq \Delta$, the central interval has length $2\eta h$ and contains at least

$$\left\lfloor \frac{2\eta h}{\Delta} \right\rfloor \geq \frac{\eta h}{\Delta}$$

grid points. Consequently,

$$W_t \geq \frac{k_0 \eta h}{\Delta}, \quad \max_u a_{u,t} \leq \frac{K_{\max} \Delta}{k_0 \eta h}, \quad \sum_u a_{u,t}^2 \leq \max_u a_{u,t}.$$

For the usual unit-interval grids, $\Delta = 1/n$ when the right endpoint is excluded, and $\Delta = 1/(n-1)$ when both endpoints are included. For $n \geq 2$, both satisfy $\Delta \leq 2/n$. Thus the single choice

$$C_K = \frac{2K_{\max}}{k_0 \eta}$$

establishes (33) for either convention, provided $\eta h \geq \Delta$. The condition $nh \geq 2/\eta$ is sufficient. If the spacing is exactly $1/n$, the sharper constant $C_K = K_{\max}/(k_0 \eta)$ and condition $nh \geq 1/\eta$ suffice. A periodic grid has n distinct locations per period; it does not count both endpoints as separate locations. There are at most $2h/\Delta + 1$ active observations, which also bounds the effective sample size from above. Both grid conventions therefore give effective sample size of order nh .

Set

$$q_n := \sqrt{\frac{C_K z_x}{nh}}, \quad A_1 := \sqrt{2}v_1 + \frac{2}{3}, \quad A_2 := \sqrt{2}v_2 + \frac{L_2}{3}. \quad (34)$$

Corollary 4.4 (A sufficient local sample size). *Under the Gaussian model, the curvature lower bound, and (33), the following condition implies (29):*

$$nh > C_K z_x \max \left\{ 1, \frac{4A_2^2}{\lambda^2}, \frac{81A_1^2 L_3^2}{\lambda^4} \right\}. \quad (35)$$

With probability at least $1 - e^{-x}$,

$$\max_t |\tilde{S}_t - S_t| \leq \frac{3A_1}{\lambda} q_n, \quad (36)$$

$$\max_t |R_t| \leq \left\{ \frac{9L_3 A_1^2}{\lambda^3} + \frac{2A_1 A_2}{\lambda^2} \right\} q_n^2. \quad (37)$$

For fixed $H > 0$, fixed σ , bounded local signal oscillation, and fixed kernel constants, λ stays bounded away from zero. The two bounds then have orders $\sqrt{(x + \log(4m))/(nh)}$ and $(x + \log(4m))/(nh)$, respectively. The smallness conditions hold eventually if $(x + \log(4m))/(nh) \rightarrow 0$. These are upper bounds on sampling error; they do not assert equality of the remainder and the squared realized error.

4.5 Population bias and recovery of a smooth signal

The preceding results control sampling error around S_t . To recover X_t , we must also bound $S_t - X_t$. Exact window symmetry cancels the linear part of the signal locally, leaving a bias controlled by its second derivative.

Assume $X_u = X(u)$, and write $X_t = X(t)$. In each window use offsets $d_{u,t} = u - t$, unwrapped locally for a periodic design. Suppose active offsets satisfy $|d_{u,t}| \leq h$. Exact window symmetry means that the weighted discrete measure $\sum_u a_{u,t} \delta_{d_{u,t}}$ is invariant under $d \mapsto -d$. This holds at equally spaced design points with an even kernel and an untruncated interior or periodic window. It need not hold at arbitrary off-grid evaluation points or at truncated boundaries.

Proposition 4.5 (Bias for symmetric windows). *Under the Gaussian model, suppose the windows have this exact symmetry and X is twice continuously differentiable on their neighborhoods, with $|X'| \leq M_1$ and $|X''| \leq M_2$. Set*

$$\mu_2(t) := \sum_u a_{u,t} d_{u,t}^2 \leq h^2, \quad \lambda_h := \frac{c}{B} e^{-2M_1^2 h^2 / B^2}. \quad (38)$$

Then

$$|S_t - X_t| \leq \frac{M_2 \mu_2(t)}{2} e^{2M_1^2 h^2 / B^2} \leq \frac{M_2 h^2}{2} e^{2M_1^2 h^2 / B^2}. \quad (39)$$

If the smallness conditions of Corollary 4.3 hold with $\lambda = \lambda_h$, its event also gives

$$\max_t |\tilde{S}_t - X_t| \leq \frac{M_2 h^2}{2} e^{2M_1^2 h^2 / B^2} + \frac{s_1}{\lambda_h} + \mathcal{R}_x, \quad (40)$$

$$\max_t |\tilde{S}_t^{\text{corr}} - X_t| \leq \frac{M_2 h^2}{2} e^{2M_1^2 h^2 / B^2} + \mathcal{R}_x. \quad (41)$$

With fixed H, σ, M_1, M_2 and regular symmetric windows, the resulting recovery bound has order

$$h^2 + \sqrt{\frac{x + \log(4m)}{nh}} + \frac{x + \log(4m)}{nh}.$$

For $z_x/n \rightarrow 0$, balancing the first two terms with $h \asymp (z_x/n)^{1/5}$ gives order $(z_x/n)^{2/5}$, with the stated high probability and for sufficiently large n to meet the sample-size condition. This is an upper bound for a fixed loss scale, not a claim of an optimal joint choice of h and H . Without exact symmetry, the range argument still gives $|S_t - X_t| \leq M_1 h$, but the displayed h^2 bound does not follow from that argument.

4.6 Scope of the conclusions

All uniform probability statements concern the finite evaluation grid. A supremum over the continuum $[0, 1]$ requires an additional argument for regularity and the size of that family of weighted processes. Adaptive choices of bandwidth or loss scale also require simultaneous control over the choices.

The correction uses the unknown population target, noise law, and curvature, so it is an oracle expansion, not an estimator computable from the observations alone. The accompanying notebook `experiments/proposition-4-1/` tests its sampling-rate prediction under Gaussian noise. Numerical rate checks do not prove the probability bound.

The general proposition allows independent non-Gaussian errors with a finite first moment. The explicit curvature and bias formulas in the Gaussian corollaries do not cover asymmetric contamination. Later IRMF components use estimated residuals that share observations; their errors are dependent and require a separate argument.

Throughout, F_t is evaluated at S_t . It cannot be replaced by a curvature evaluated at X_t without bounding their difference.

A Proofs

The argument has two stages. Lemma 4.2 controls the random score and curvature at the population target. Conditional on that event, the proof of Proposition 4.1 is deterministic: localize the empirical minimizer, apply scalar Taylor expansion, and replace empirical curvature by population curvature. The remaining proofs specialize these bounds and control the population bias.

A.1 Score and curvature concentration

Proof of Lemma 4.2. Differentiating (16) gives

$$\begin{aligned} f'(z) &= - \sum_u \phi'(X_u - z)w_{u,t}, & f'_o(z) &= - \sum_u \rho'(Y_u - z)w_{u,t}, \\ f''(z) &= \sum_u \phi''(X_u - z)w_{u,t}, & f''_o(z) &= \sum_u \rho''(Y_u - z)w_{u,t}. \end{aligned}$$

Since differentiation under the expectation is justified by (3), $\phi^{(j)}(a) = \mathbb{E}\rho^{(j)}(a + \varepsilon)$ for $j = 1, 2, 3$. The identities in part (i) follow.

For the first identity, set

$$Z_{u,t} := w_{u,t} \left\{ \rho'(Y_u - x^*) - \mathbb{E}\rho'(Y_u - x^*) \right\}.$$

These variables are independent and centered. Moreover,

$$|Z_{u,t}| \leq 2L_1 k_t, \quad \sum_u \mathbb{E} Z_{u,t}^2 \leq L_1^2 Q_t^2.$$

Bernstein's inequality therefore yields, for every $z > 0$,

$$\mathbb{P} \left(\left| \sum_u Z_{u,t} \right| > L_1 \left\{ Q_t \sqrt{2z} + \frac{2k_t z}{3} \right\} \right) \leq 2e^{-z}. \quad (42)$$

For the second identity, use

$$\tilde{Z}_{u,t} := w_{u,t} \left\{ \rho''(Y_u - x^*) - \mathbb{E}\rho''(Y_u - x^*) \right\}.$$

Convexity and (3) give

$$|\tilde{Z}_{u,t}| \leq L_2 k_t, \quad \sum_u \mathbb{E} \tilde{Z}_{u,t}^2 \leq L_2^2 Q_t^2.$$

Hence

$$\mathbb{P} \left(\left| \sum_u \tilde{Z}_{u,t} \right| > L_2 \left\{ Q_t \sqrt{2z} + \frac{k_t z}{3} \right\} \right) \leq 2e^{-z}. \quad (43)$$

Take $z = z_x = x + \log(4m)$ in (42) and (43), and take the union over the two inequalities and over $t \in \mathcal{T}_n$. The exceptional probability is at most $4me^{-z_x} = e^{-x}$. Equations (9) and (10) now prove part (i).

Part (ii) is deterministic. By (3) and (11),

$$\sup_{z \in \mathbb{R}} \frac{1}{D^2} \left| \sum_u \rho'''(Y_u - z)w_{u,t} \right| \leq \frac{L_3 W_t}{D^2} \leq C_3.$$

Finally, $|\phi'''| \leq L_3$. The mean value theorem and $|z - x^*| \leq r/D$ for $z \in U_t(r)$ give

$$\begin{aligned} \frac{1}{D^2} \left| \sum_u \{ \phi''(X_u - z) - \phi''(X_u - x^*) \} w_{u,t} \right| \\ \leq \frac{L_3 W_t}{D^2} |z - x^*| \leq \frac{C_3 r}{D}, \end{aligned}$$

which proves part (iii). $\square$

A.2 Scalar localization and the remainder

Proof of Proposition 4.1. Fix $t \in \mathcal{T}_n$. Since $f'(x^*) = 0$,

$$f'_\circ(x^*) = - \sum_u \{ \rho'(Y_u - x^*) - \mathbb{E} \rho'(Y_u - x^*) \} w_{u,t} = -\nabla_t. \quad (44)$$

Lemma 4.2(i) also gives

$$\frac{|F_\circ - F|}{F} \leq \frac{C_2}{D} \leq \delta. \quad (45)$$

In particular, $F_\circ \geq (1 - \delta)F > 0$. With

$$b_\circ := \frac{D}{F_\circ} |f'_\circ(x^*)|, \quad (46)$$

the score bound and (45) imply

$$b_\circ \leq \frac{C_1}{1 - \delta}. \quad (47)$$

Use the scaled coordinate $v = D(z - x^*)$, and put

$$g(v) := f_\circ(x^* + v/D), \quad a := g''(0) = F_\circ/D^2, \quad \tau := C_3/D. \quad (48)$$

Then $a \geq 1 - \delta$, $|g'''(v)| \leq \tau$, and $b_\circ = |g'(0)|/a$. Consequently,

$$\frac{\tau b_\circ}{a} \leq \frac{C_1 C_3}{D(1 - \delta)^2} < \frac{4}{9}. \quad (49)$$

If $b_\circ = 0$, convexity and $g''(0) > 0$ imply that 0 is the unique minimizer, and the claimed bounds are immediate. Otherwise set $r = 3b_\circ/2$. Taylor expansion of g' at zero yields

$$|g'(v) - g'(0) - av| \leq \frac{\tau}{2} v^2.$$

Since $|g'(0)| = ab_\circ$,

$$\begin{aligned} g'(r) &\geq -ab_\circ + ar - \frac{\tau r^2}{2} = \frac{ab_\circ}{2} - \frac{9\tau b_\circ^2}{8} > 0, \\ g'(-r) &\leq ab_\circ - ar + \frac{\tau r^2}{2} = -\frac{ab_\circ}{2} + \frac{9\tau b_\circ^2}{8} < 0. \end{aligned}$$

Moreover, for $|v| \leq r$,

$$g''(v) \geq a - \tau r > \frac{a}{3} > 0.$$

There is therefore a unique zero $v_\circ \in (-r, r)$ of g' . Convexity on $\mathbb{R}$ makes it the unique global minimizer: the derivative is negative to the left of this interval and positive to its right. This proves

$$|v_\circ| = D|x_\circ - x^*| \leq \frac{3}{2}b_\circ \leq \frac{3C_1}{2(1 - \delta)}. \quad (50)$$

At the zero of g' , the same Taylor expansion gives

$$\left| a \left(v_o + \frac{g'(0)}{a} \right) \right| \leq \frac{\tau}{2} v_o^2 \leq \frac{9\tau}{8} b_o^2. \quad (51)$$

Equivalently, the left-hand side is $D^{-1}F_o|x_o - x^* + F_o^{-1}f'_o(x^*)|$. Using (45) and (47), we obtain

$$|x_o - x^* + F_o^{-1}f'_o(x^*)| \leq \frac{9C_1^2C_3}{8D^2(1-\delta)^3}. \quad (52)$$

It remains to replace F_o by the population curvature F . By (45), (19), and $f'(x^*) = 0$,

$$\begin{aligned} |F_o^{-1}f'_o(x^*) - F^{-1}f'_o(x^*)| &= \frac{|F_o - F|}{F_o F} |f'_o(x^*)| \\ &\leq \frac{C_1 C_2}{D^2(1-\delta)}. \end{aligned} \quad (53)$$

The triangle inequality applied to (52) and (53), followed by (44), proves

$$D|x_o - x^* - F^{-1}\nabla_t| \leq \frac{1}{D} \left\{ \frac{9C_1^2C_3}{8(1-\delta)^3} + \frac{C_1C_2}{1-\delta} \right\}. \quad (54)$$

Taking the supremum over $t \in \mathcal{T}_n$ and using $D_t \geq D_{\min}$ gives (13). $\square$

Remark (The remainder coefficient). *The coefficient $9/8$ is $\frac{1}{2}(\frac{3}{2})^2$, from Taylor's theorem and localization. The earlier completion used $3/4$ from Theorem A.9 of the IMF draft. That generic coefficient fails even in one dimension. On $(-1, 0.9)$, let*

$$q(v) = -0.4v + \frac{v^2}{2} - \frac{v^3}{6}.$$

This function is strongly convex, $q''(0) = 1$, $|q'''| = 1$, and its Newton step from zero is $b = 0.4 < 4/9$. The ball of radius $3b/2 = 0.6$ lies inside the domain. The minimizer is $v_ = 1 - \sqrt{0.2}$, but*

$$|v_* - b| \approx 0.152786 > \frac{3}{4}b^2 = 0.12.$$

For a globally strongly convex example on $\mathbb{R}$, retain $q''(v) = 1 - v$ for $v \leq 0.7$, and set $q''(v) = 0.2 + 0.1e^{-10(v-0.7)}$ for $v > 0.7$, integrating with the same $q(0)$ and $q'(0)$. This is a C^3 extension with $q'' \geq 0.2$ and $|q'''| \leq 1$; it has the same minimizer and remainder. The present proof does not invoke the appendix theorem.

A.3 Gaussian constants

Proof of Corollary 4.3. The identity (25) follows because $\varepsilon + HZ \sim N(0, B^2)$; the target and curvature formulas follow by differentiation. For a Lipschitz function G and an independent copy ε' of ε ,

$$\text{Var}\{G(X_u + \varepsilon)\} = \frac{1}{2}\mathbb{E}\{G(X_u + \varepsilon) - G(X_u + \varepsilon')\}^2 \leq \text{Lip}(G)^2\sigma^2.$$

Apply this with $G(y) = \rho'_H(y - S_t)$ and $G(y) = \rho''_H(y - S_t)$. Their Lipschitz constants are at most L_2 and L_3 . Combining this estimate with the boundedness estimates from Lemma 4.2 gives variance bounds v_1^2 and v_2^2 . Bernstein's inequality and the same union bound therefore give an event of probability at least $1 - e^{-x}$ on which

$$\max_t |\bar{\nabla}_t| \leq s_1, \quad \max_t |\bar{F}_{o,t} - \bar{F}_t| \leq s_2, \quad \bar{F}_{o,t} = \sum_u a_{u,t} \rho''_H(Y_u - S_t).$$

For the normalized objectives, the deterministic proof of Proposition 4.1 applies with the bounds

$$C_1 = \frac{s_1}{\sqrt{\lambda}}, \quad C_2 = \frac{s_2}{\sqrt{\lambda}}, \quad C_3 = \frac{L_3}{\lambda}, \quad D_{\min} \geq \sqrt{\lambda}, \quad \delta \leq d.$$

Condition (29) is sufficient for (12). Substitution in (14) and (54), followed by division by $D_t \geq \sqrt{\lambda}$, proves the raw and remainder bounds. The score bound and $\bar{F}_t \geq \lambda$ give (31). $\square$

A.4 Sufficient local sample size

Proof of Corollary 4.4. The mass bounds imply

$$s_1 \leq \sqrt{2}v_1q_n + \frac{2}{3}q_n^2, \quad s_2 \leq \sqrt{2}v_2q_n + \frac{L_2}{3}q_n^2.$$

Condition (35) implies

$$q_n < \min \left\{ 1, \frac{\lambda}{2A_2}, \frac{\lambda^2}{9A_1L_3} \right\}.$$

Therefore $s_1 \leq A_1q_n$, $s_2 \leq A_2q_n$, $d < 1/2$, and

$$\frac{s_1L_3}{\lambda^2(1-d)^2} \leq \frac{4A_1q_nL_3}{\lambda^2} < \frac{4}{9}.$$

Apply Corollary 4.3 and use $(1-d)^{-1} \leq 2$ and $(1-d)^{-3} \leq 8$. $\square$

A.5 Bias for symmetric windows

Proof of Proposition 4.5. Write $\psi_B = \rho'_B$ and $G_t(z) = \sum_u a_{u,t}\psi_B(X_u - z)$, so $G_t(S_t) = 0$. Oddness of ψ_B and exact symmetry give

$$\sum_u a_{u,t}\psi_B(X'_t(t)d_{u,t}) = 0.$$

Taylor's theorem for the signal and $\sup_r \psi'_B(r) = c/B$ imply

$$\begin{aligned} |G_t(X_t)| &\leq \frac{c}{B} \sum_u a_{u,t} |X(u) - X(t) - X'_t(t)d_{u,t}| \\ &\leq \frac{cM_2}{2B} \mu_2(t). \end{aligned}$$

Since S_t lies in the range of the active signal values, $|S_t - X_t| \leq M_1h$. For any z between S_t and X_t , $|X_u - z| \leq 2M_1h$ for every active u . It follows that

$$-G'_t(z) = \frac{c}{B} \sum_u a_{u,t} e^{-(X_u - z)^2/(2B^2)} \geq \lambda_h.$$

The mean value theorem now gives $|S_t - X_t| \leq |G_t(X_t)|/\lambda_h$, which proves (39). The same derivative bound at S_t proves $\bar{F}_t \geq \lambda_h$. Finally, use the exact decomposition

$$\tilde{S}_t - X_t = (S_t - X_t) + F_t^{-1}\nabla_t + R_t$$

and Corollary 4.3. Subtracting the leading term gives the oracle-corrected bound. $\square$